SPACE FACULTY

Let’s learn about **satellites**!

Space mission and **satellite** design course

How do we do satellites?

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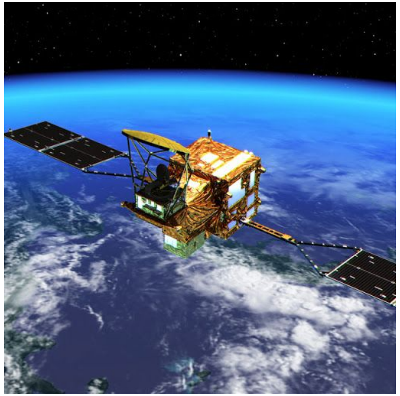
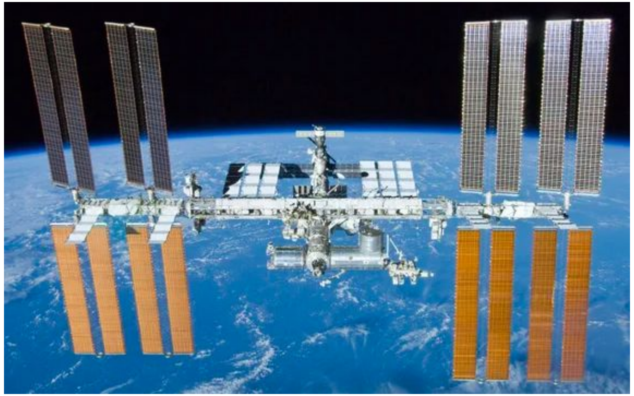
How is made a satellite?

A satellite is a **semi-autonomous** system able to perform operations.

In order to do so, each satellite is composed of a set of key subsystems with specific functions





Satellite sub-systems

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Satellite sub-systems

- Structure
- Power system (batteries, recharge system, power distribution)
- Communication system (transceiver, antennas)
- Computing system (OBC)
- Sensors (Magnetometer, Gyroscope, etc.)
- Navigation systems (GPS, Glonass, Galileo)
- Payload (Camera, sensor, etc)
- On-board software
- Thermal control system
- Physical and SW interfaces
- Attitude control system
- Propulsion system

Satellite design

- Satellite design involves multiple engineering disciplines:
    - Mechanical engineering
    - Electronic engineering
    - Software engineering
    - Telecommunication engineering
    - Chemicals and material engineering
    - Astrophysics and math
    - and also....Aerospace engineering!
- It is not strictly necessary to be aerospace engineer to work on satellite industry

Structure

Structural system is one of the primary subsystem with the scope to:

- Contain other subsystems
- Protect from mechanical loads (Launch)
- Distribute thermal and electrostatic loads

Power System

PMDU - Power Management & Distribution Unit has the scope to provide electrical power to all the platform

It is composed by:

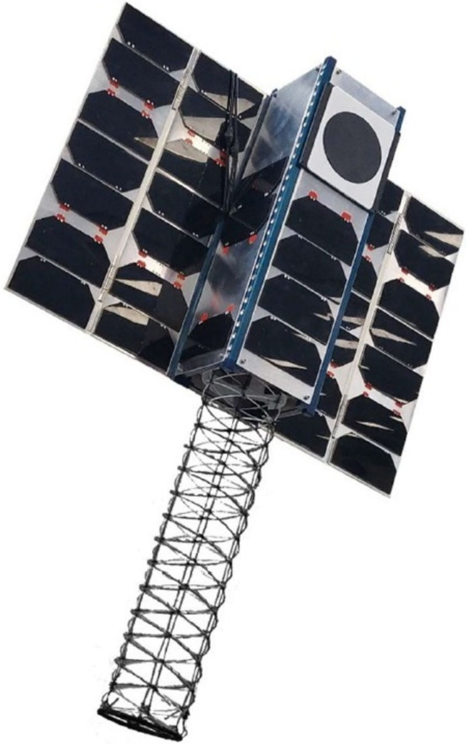
- Power generation systems
- Energy storage systems
- Power management and distribution systems

Communication system

- Communication system has the scope to allow radio (not only) communication between the satellite and an earth station.

- Thanks to communication system we can download data from satellite and send commands to control it.

- Typically communication system is composed by:
  TX/RX Module(s)
  TX/RX Antenna(s)



https://www.esa.int/ESA_Multimedia/Images/2020/11/CubeSat_with_helical_antenna

Communication system

In some cases we have more than one communication system on board:

- **TT&C system:**
  Communication system dedicated to satellite telemetry data and low data rates

- **Payload com system:**
  Communication system dedicated to payload data; typically is high performance communication system with high data rate.

On-board computer

- The OBC is the operational core of the satellite with the scope to control and supervise all the satellite subsystems

- It is the computer of the satellite's avionic sub-system, i.e. the unit where the **On Board Software** run.

- The main layer of an OBC is the microprocessor board, consisting of microprocessor, non-volatile memories, volatile memories and the companion chip that connects the microprocessor to different peripherals.

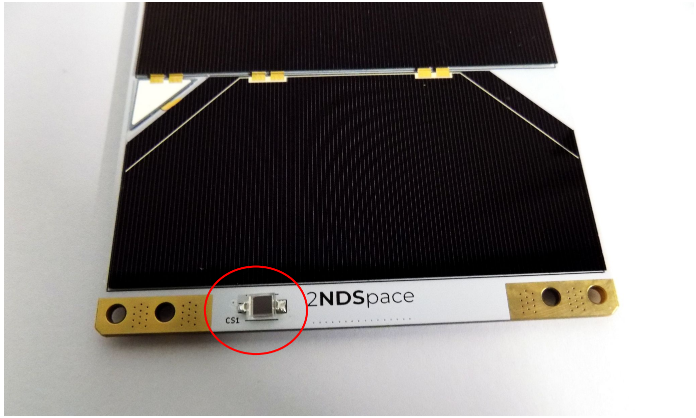
On-board software

- On-board software is the main software implementing the satellite's vital functions such as: attitude and orbit control in both nominal and non-nominal cases, tele-commands execution or dispatching, housekeeping telemetry gathering and formatting, on board time synchronisation and distribution, failure detection, isolation and recovery, etc.

https://www.esa.int/Enabling_Support/Space_Engineering_Technology/Onboard_Computers_and_Data_Handling/Onboard_Computers

Sensors

- Attitude determination system includes all the sensors that allow to determine the attitude of the satellites and its orientation in space:
    - Gyroscopes
    - Accelerometer
    - Magnetometer
    - Star trackers
    - Sun/Earth sensors
    - Thermal Sensors



- Most of sensors require to be integrated on the satellite external surface and in some cases to be placed far from the sat body

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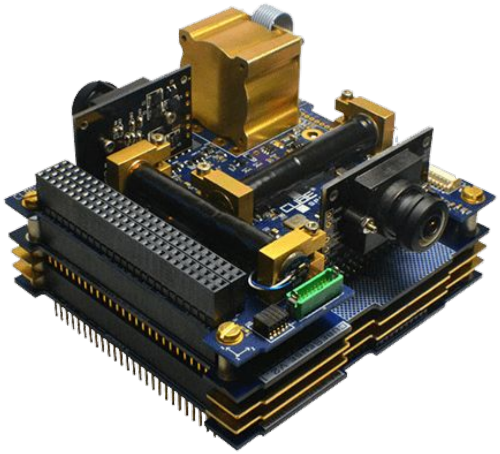
Attitude control system

- Attitude control system includes all the actuators allowing to change the **attitude** of the satellite with respect to its inertial axes.
- Together with attitude determination system it compose the **ADCS subsystem** (eg. CubeADCS from Cubespace)
    - Magnetic actuator
    - Reaction wheels
    - Cold gas propulsion



https://www.cubesatshop.com/product-category/integrated-adcs/

- ADCS actuators does not change the orbit of the satellite or the position of the satellite over its orbit, they just allow to control its orientation

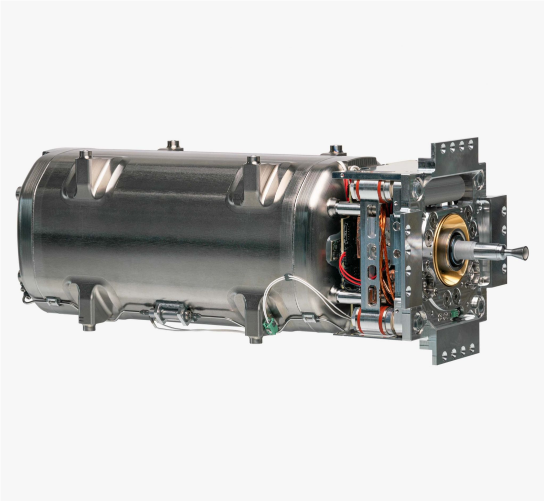
Navigation

- Navigation systems include systems that allows to define the position in space of the satellite

- Typically it is used global navigation system based on GPS, Glonass, Galileo, etc.

Propulsion system

- Propulsion systems allows to move the satellite along the orbit or to effectively change the orbit.
- There are several types of propulsion systems according to the required scope:
    - Orbital transfer
    - Station keeping
    - De-orbiting



https://nanoavionics.com/cubesat-components/cubesat-propulsion-system-epss/

Thermal control system

- Thermal control system is required to control the heat exchange and thermal stability of the satellite.
- Most of electronic subsystem requires to operate in a specific temperature range, moreover subsystem are altered/damaged by the thermal stresses.
- Thermal control system can be active (really complex!) through the use of refrigerant fluids circulating through pumps and active systems.
- Passive systems for thermal control implies the use of radiator and designed heat path

Practical aspects

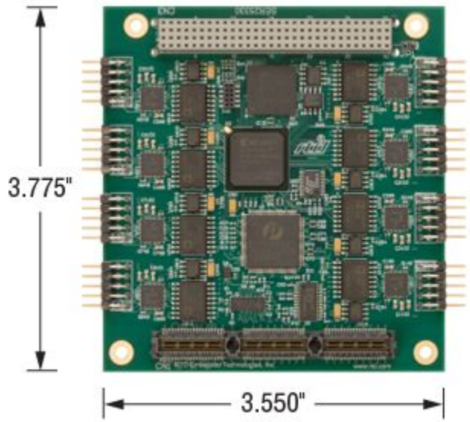
- Some subsystems requires to be integrated inside the satellite:

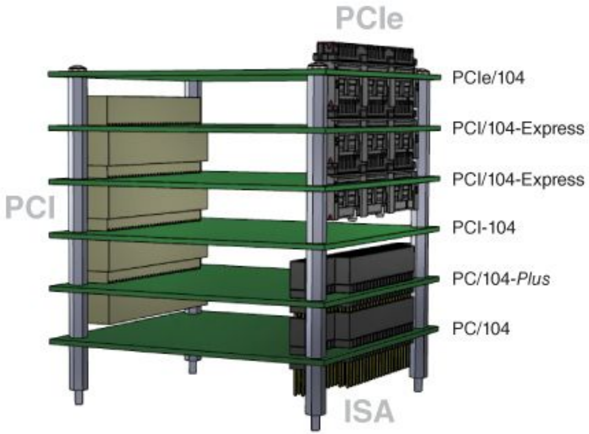
- Other subsystems ***require*** to be integrated on the external surface of the satellite:
    - Solar Panels
    - Antennas
    - Sensors (Sun sensors, magnetometer, thermal sensors)

- Externally mounted subsystem has to face the issue of **harsch space** environment

Practical aspects

- Internal components can be integrated in different ways.
- In cubesat application it is quite exploited **PC104** standard for subsystem design and integration.





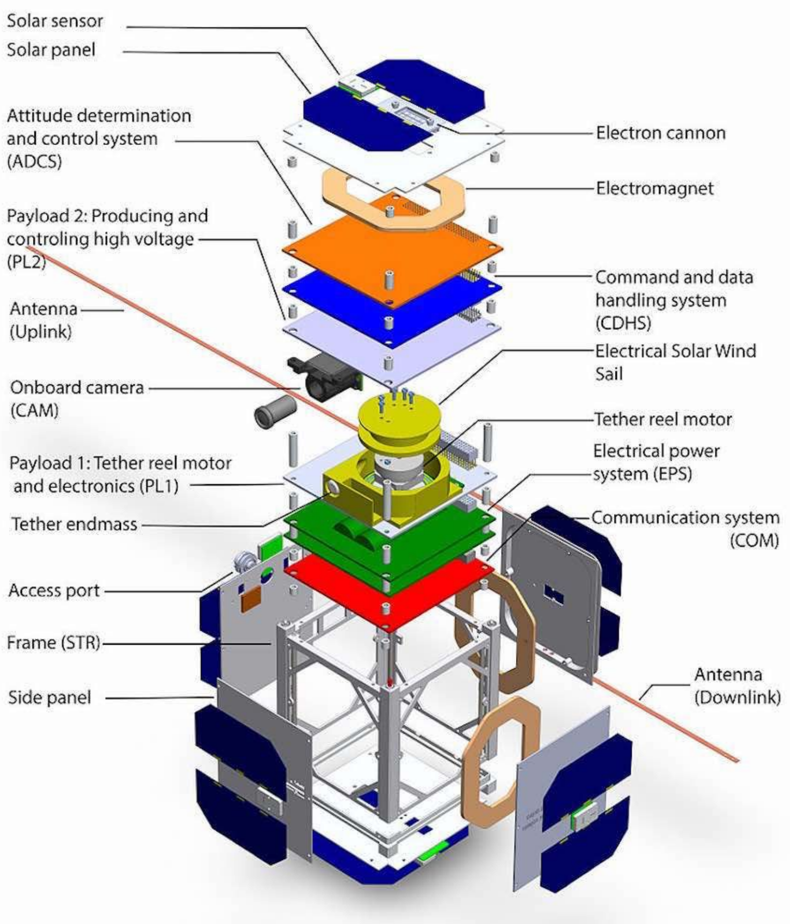
https://www.rtd.com/PC104/

Practical aspects

- Satellite subsystem shall be accessible from outside for programming, monitoring, battery charging.
- In order to do so, Umbilical connections shall be taken into account. These consists in exposed connectors that can be accessed when the satellite is fully integrated.
- In order to be correctly operated and configured, most subsystem might need supporting equipment typically represented by MGSE and EGSE.

https://www.rtd.com/PC104/

Cubesat examples



The structure of cubesat ESTCube-1

https://www.fictiv.com/articles/satellite-101-what-is-a-cubesat

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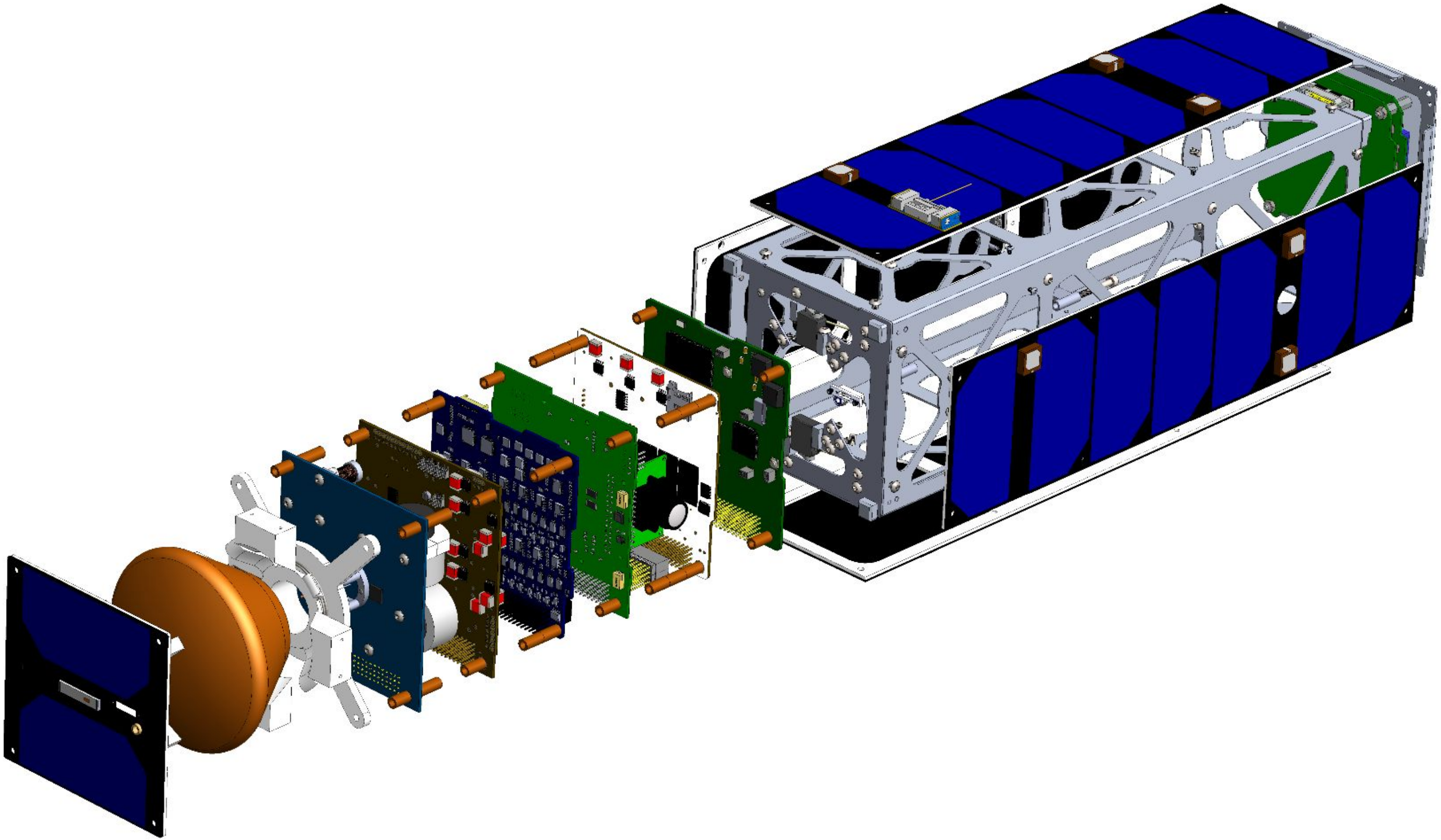


Satellite design software

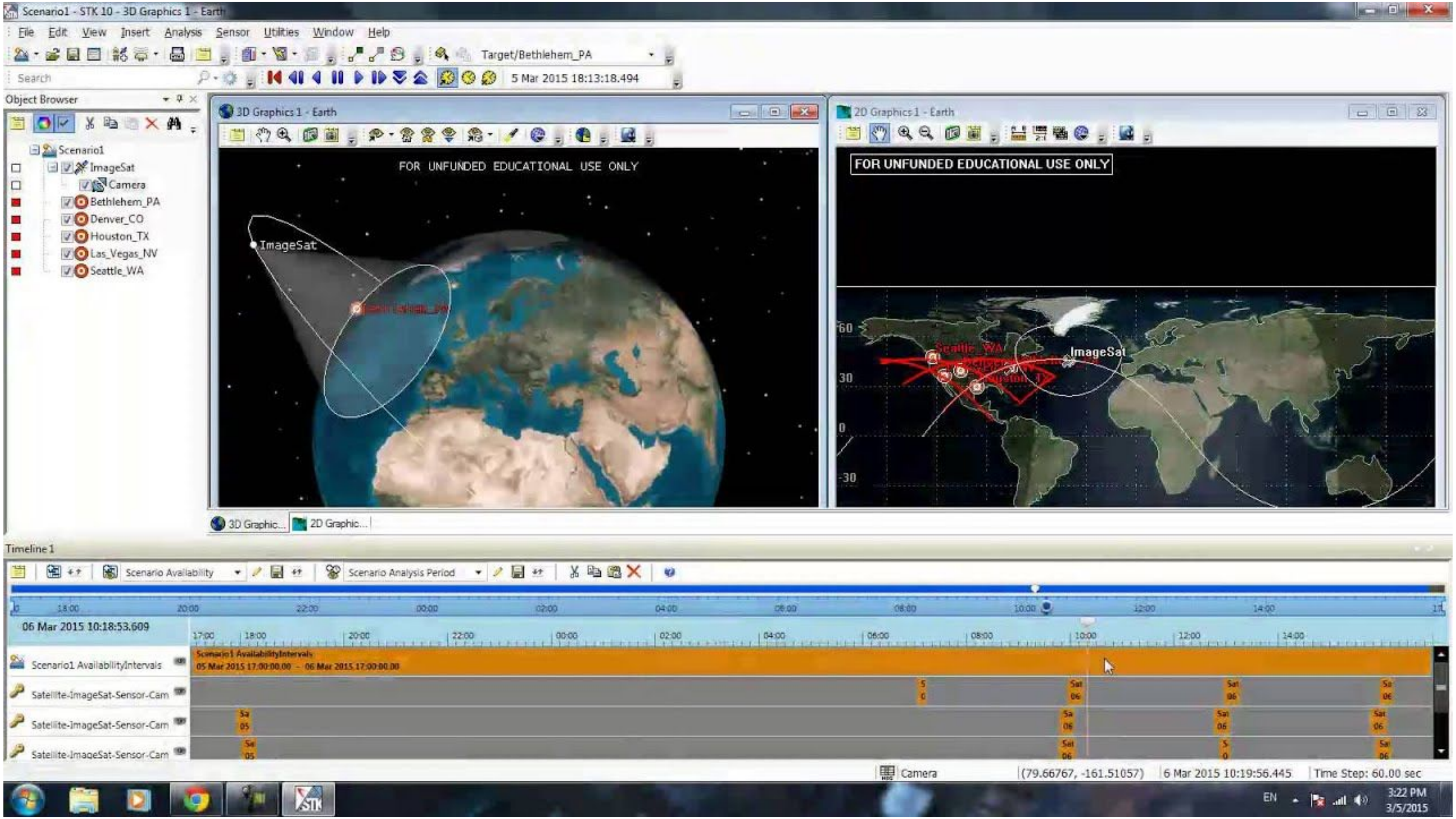
In a complete mission and platform design, a wide range of software can be exploited:

- **Mission analysis softwares:** tools for designing the overall mission, the orbit analysis and satellite lifetime simulation
- **Engineering design softwares:** tools for low level design and performances analysis and verification the overall mission, the orbit analysis and satellite lifetime simulation
- **Project Management softwares:** tools for managing complete projects from managerial, costing and schedule point of view.

Mission analysis

STK (System tool kit) from AGI is a multi-physics software application from Analytical Graphics, Inc. that enables engineers and scientists to perform complex analyses of ground, sea, air, and space platforms, and to share results in one integrated environment.

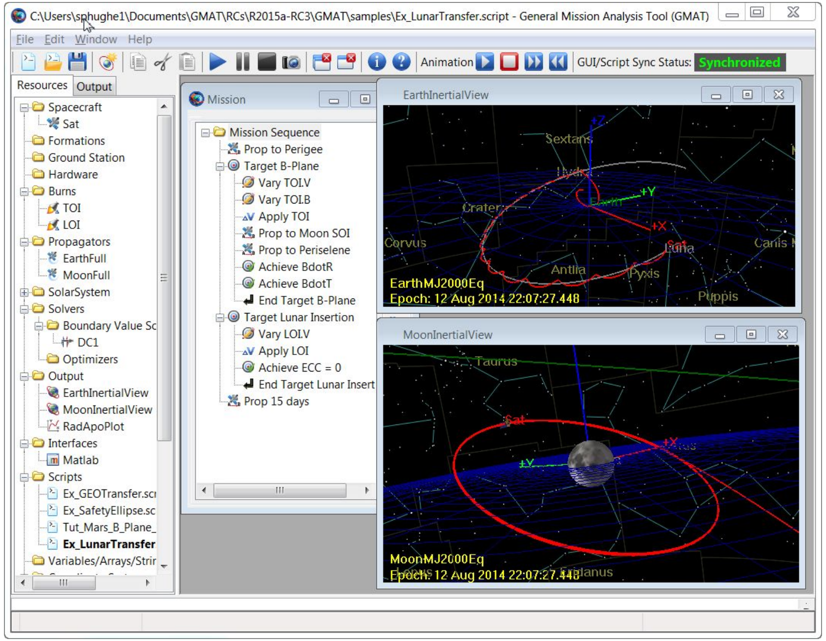
- Mission design (number of satellites, type, orbital distribution)
- Ground station access (number of accesses, lik type and duration, etc)
- Orbital coverage analysis (ground target definition, accesses, revisit time, etc)
- Mission propagation in time (orbital evolution, performances analysis, etc)
- And a lot more...



https://www.youtube.com/watch?app=desktop&v=-qvRY0G08N0

Mission analysis - GMAT

Another tool certified for mission analysis is **GMAT**, or General Mission Analysis Tool. It is developed by NASA and it is completely free and open source. It has been used on a variety of space missions and for some of the simulations you have seen in lesson 2!



Engineering

Engineering design softwares allow to design parts, assemblies, softwares, algorithms, etc.. starting from scratch to final product.

- **Mechanical design**
- **Electrical design**
- **Simulation and computation**
- **Analysis and verification**
- **Software design (coding)**

Nowadays most of software are have increasingly expanded their scope of operations, becoming extremely flexible to cover more than one of the aspects indicated above.

We will have an overview based on “it is mostly known for…”

.

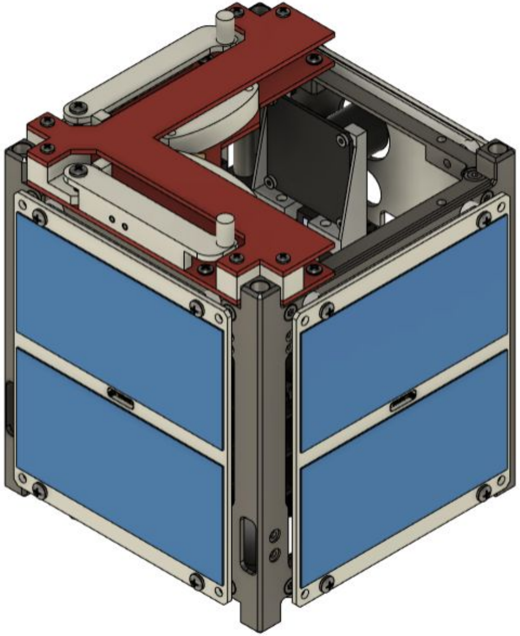
Mechanical Design

Engineering design softwares allow to design parts, assemblies, softwares, algorithms, etc.. starting from scratch to final product.

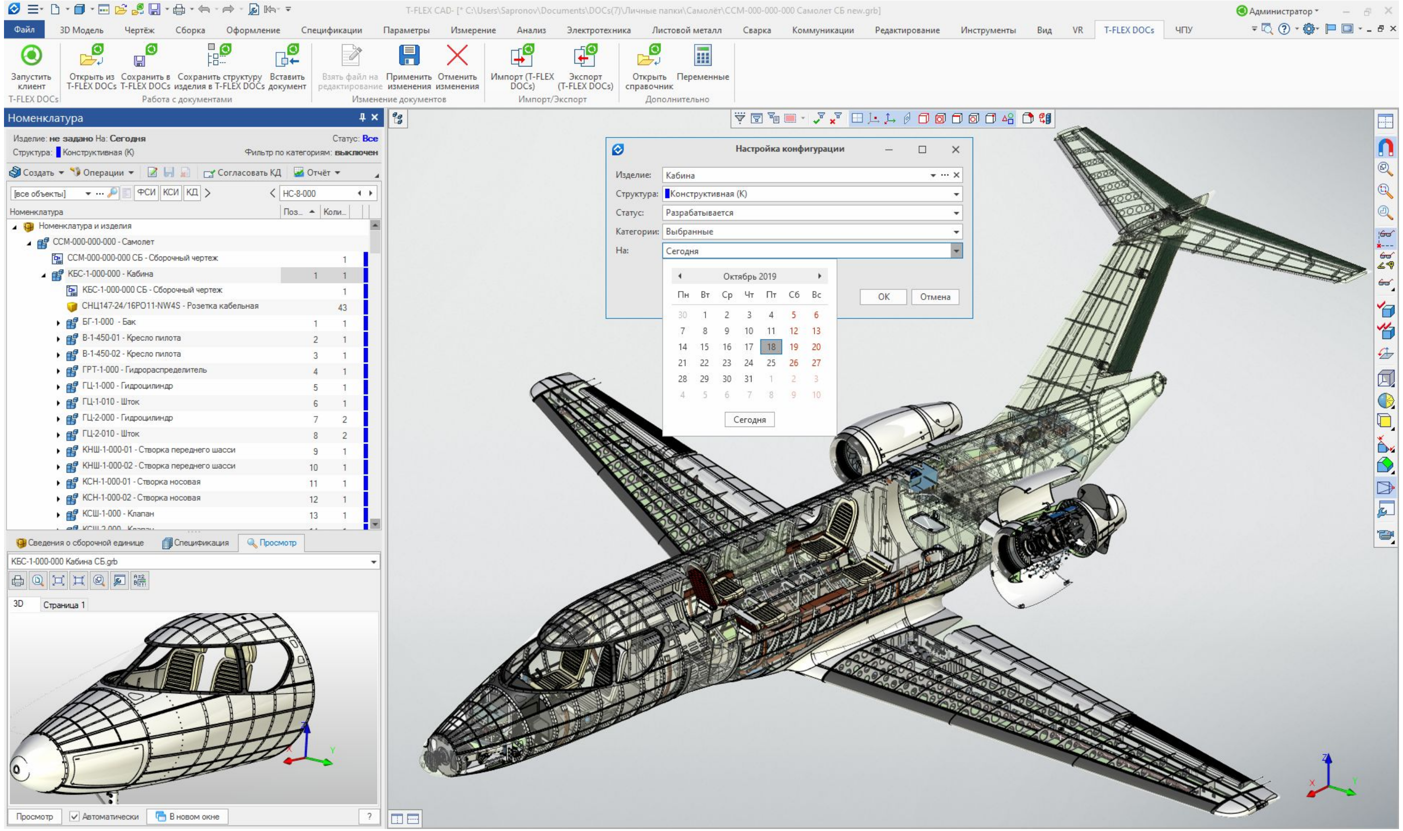
- **Mechanical design**

Allows 3D modelling of parts and systems assembly allowing to generate production files. By these softwares can be maintained under control the **overall mechanical and physical layout** of the satellite by measuring and analyzing inertial properties, defining assembly process and evaluating system interferences.

Some examples: CATIA, Solidworks, PTC CREO, Fusion 360, etc...



Mechanical Design



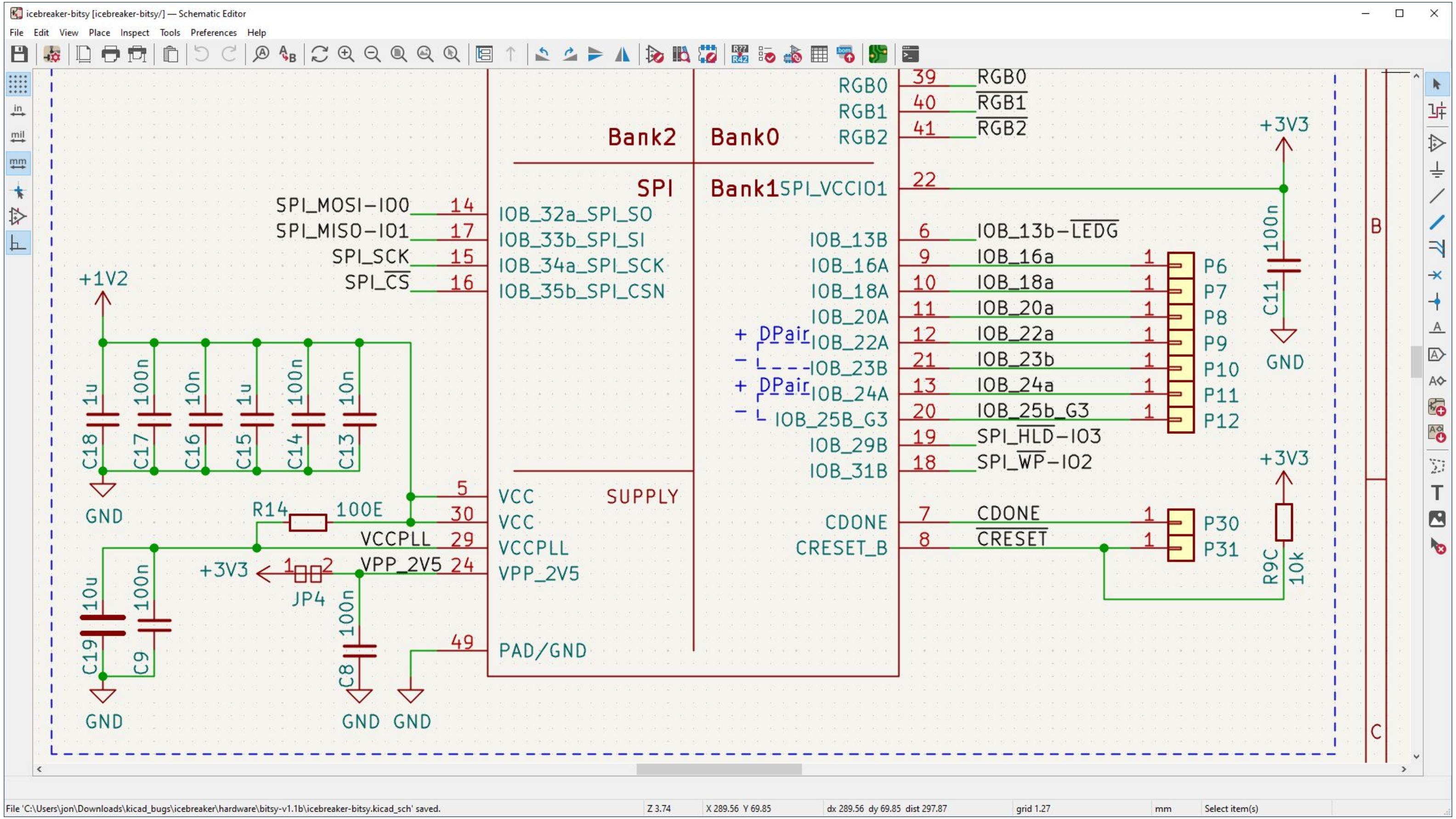
Electrical design

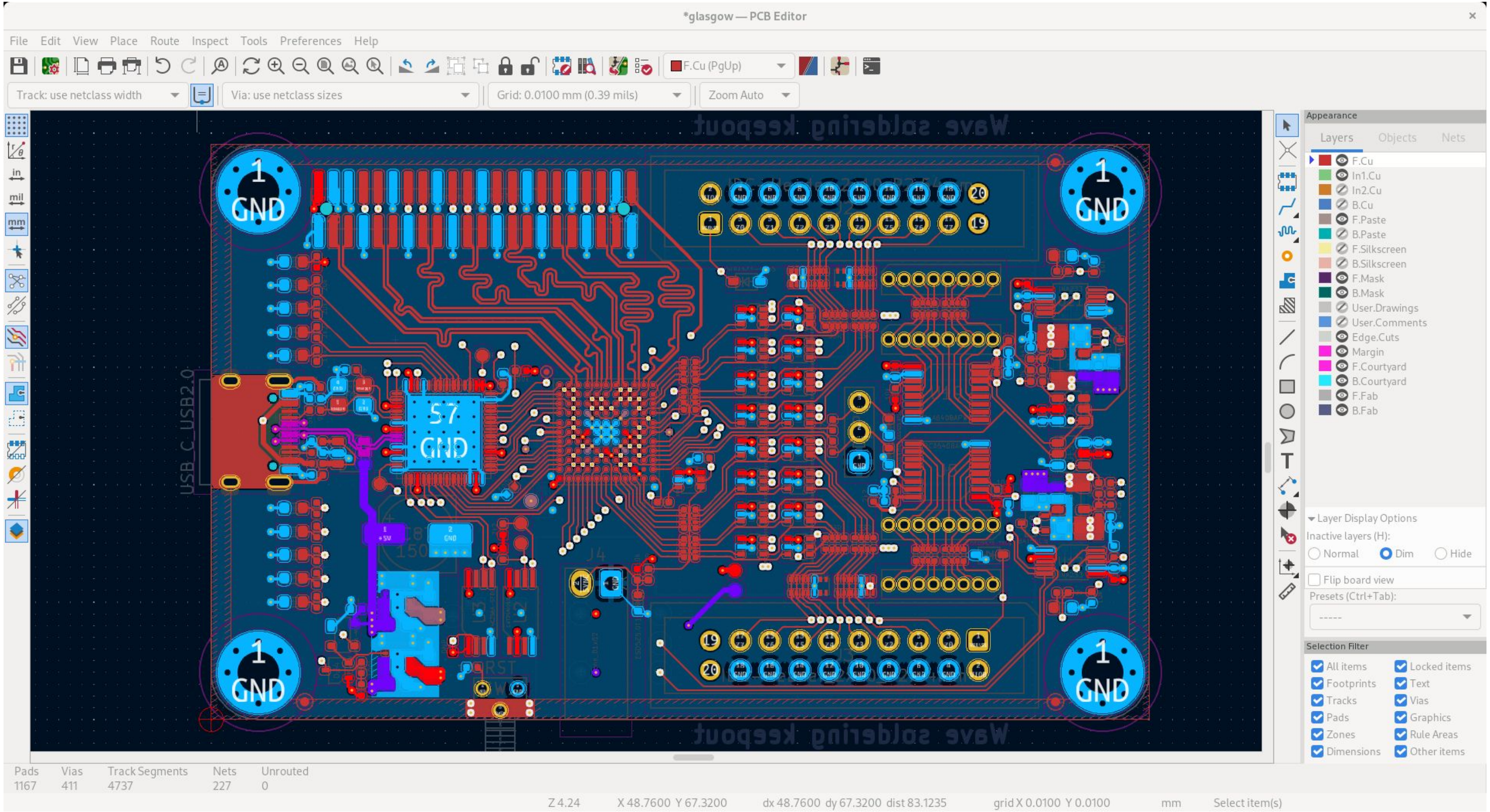
Engineering design softwares allow to design parts, assemblies, softwares, algorithms, etc.. starting from scratch to final product.

- **Electrical Design**

Allows design of electrical schematics and printed circuit boards. By this softwares it is also possible to generate production files and 3D model of electrical PCB to integrate in mechanical design softwares.

Some examples: Eagle (Fusion360), Kicad, Altium, etc..


icebreaker-bitsy [icebreaker-bitsy/] — Schematic Editor
File Edit View Place Inspect Tools Preferences Help
Bank2
Bank0
SPI
Bank1
SUPPLY
SPI_MOSI-IO0 14 IOB_32a_SPI_SO
SPI_MISO-IO1 17 IOB_33b_SPI_SI
SPI_SCK 15 IOB_34a_SPI_SCK
SPI_CS 16 IOB_35b_SPI_CSN
RGB0 39 RGB0
RGB1 40 RGB1
RGB2 41 RGB2
SPI_VCCIO1 22
IOB_13B 6 IOB_13b-LEDG
IOB_16A 9 IOB_16a 1 P6
IOB_18A 10 IOB_18a 1 P7
IOB_20A 11 IOB_20a 1 P8
IOB_22A 12 IOB_22a 1 P9
IOB_23B 21 IOB_23b 1 P10
IOB_24A 13 IOB_24a 1 P11
IOB_25B_G3 20 IOB_25b_G3 1 P12
IOB_29B 19 SPI_HLD-IO3
IOB_31B 18 SPI_WP-IO2
DPair
DPair
CDONE 7 CDONE 1 P30
CRESET_B 8 CRESET 1 P31
+1V2
1u 100n 10n 1u 100n 10n
C18 C17 C16 C15 C14 C13
GND
5 VCC
R14 100E 30 VCC
VCCPLL 29 VCCPLL
+3V3 JP4 VPP_2V5 24 VPP_2V5
10u 100n 100n
C19 C9 C8
49 PAD/GND
GND GND GND
+3V3
100n C11
GND
+3V3
R9C 10k
B
C
File 'C:\Users\jon\Downloads\kicad_bugs\icebreaker\hardware\bitsy-v1.1b\icebreaker-bitsy.kicad_sch' saved.
Z 3.74 X 289.56 Y 69.85 dx 289.56 dy 69.85 dist 297.87 grid 1.27 mm Select item(s)


*glasgow — PCB Editor
File Edit View Place Route Inspect Tools Preferences Help
F.Cu (PgUp)
Track: use netclass width
Via: use netclass sizes
Grid: 0.0100 mm (0.39 mils)
Zoom Auto
Wave soldering keepout
USB_C_USB2.0
Appearance
Layers Objects Nets
F.Cu
In1.Cu
In2.Cu
B.Cu
F.Paste
B.Paste
F.Silkscreen
B.Silkscreen
F.Mask
B.Mask
User.Drawings
User.Comments
Edge.Cuts
Margin
F.Courtyard
B.Courtyard
F.Fab
B.Fab
Layer Display Options
Inactive layers (H):
Normal Dim Hide
Flip board view
Presets (Ctrl+Tab):
Selection Filter
All items Locked items
Footprints Text
Tracks Vias
Pads Graphics
Zones Rule Areas
Dimensions Other items
Pads 1167
Vias 411
Track Segments 4737
Nets 227
Unrouted 0
Z 4.24
X 48.7600 Y 67.3200
dx 48.7600 dy 67.3200 dist 83.1235
grid X 0.0100 Y 0.0100
mm
Select item(s)

Simulation and computation

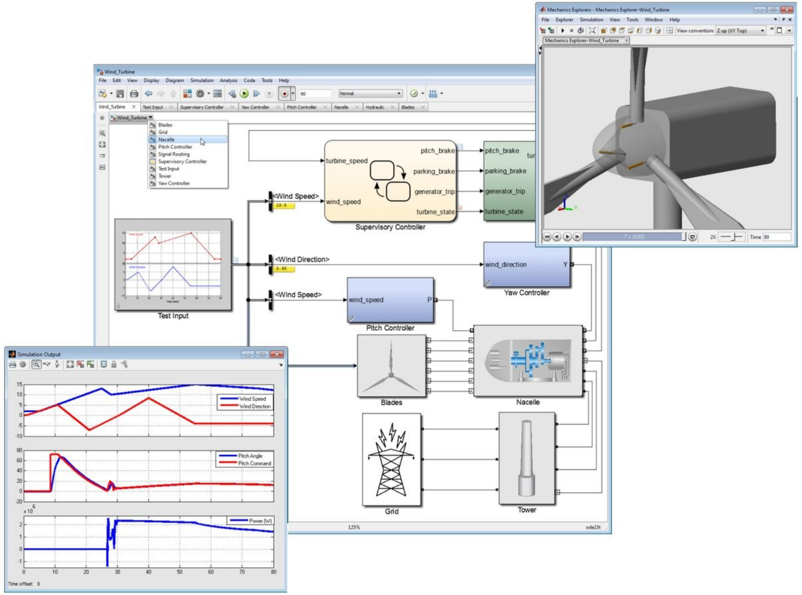
- Typically exploited computational softwares includes microsoft excel and Matlab.

- MATLAB is a proprietary multi-paradigm programming language and numeric computing environment developed by MathWorks ana allowing matrix manipulations, plotting of functions and data, implementation of algorithms, creation of user interfaces, and interfacing with programs written in other languages.

- MATLAB can be exploited for a wide range of satellite mission/subsystem analysis, design, and simulations....

Simulation and computation

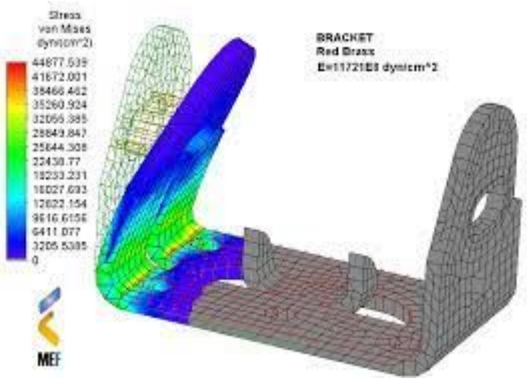
- Simulink is a MATLAB-based graphical programming environment for modeling, simulating and analyzing multidomain dynamical systems. Its primary interface is a graphical block diagramming tool and a customizable set of block libraries.

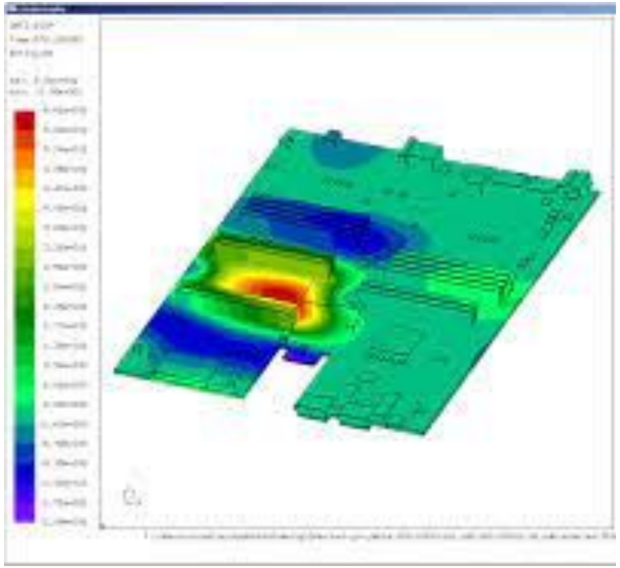


Analysis

- Analysis softwares allow to simulate mechanical, electrical, thermal operations in order to verify design compliance before production and integration.

- Mechanical Analysis (FEM) (ANSYS, NASTRAN, etc..)
- Electricals simulation (Proteus, etc..)
- Thermal Analysis (ESATAN, Thermal Desktop, etc)
- RF Analysis and simulation (CENOS, Keysights, etc..)

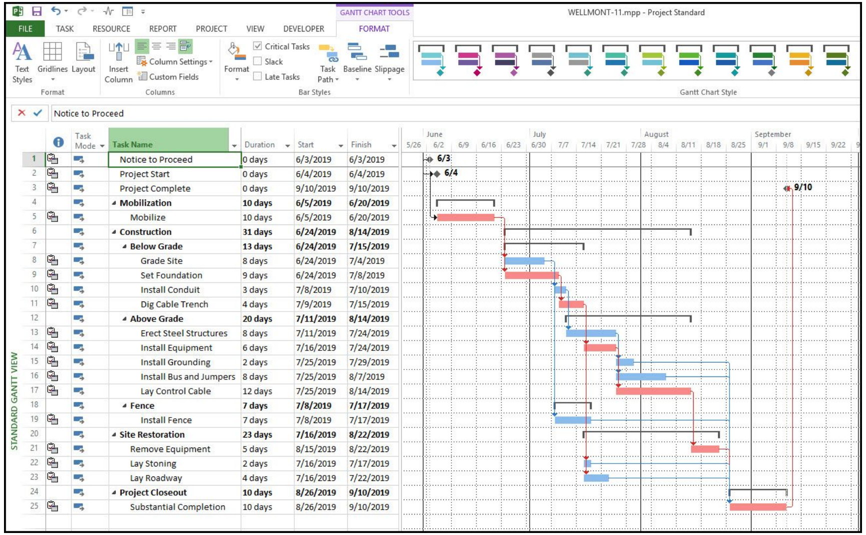




Project management

- These softwares are fundamental to correctly handle a project in terms of deliverable, resources allocation, scheduling and economic sustainability.

- Microsoft project is a well known project management software



	Task Name	Duration	Start	Finish
1	Notice to Proceed	0 days	6/3/2019	6/3/2019
2	Project Start	0 days	6/4/2019	6/4/2019
3	Project Complete	0 days	9/10/2019	9/10/2019
4	**Mobilization**	**10 days**	**6/5/2019**	**6/20/2019**
5	Mobilize	10 days	6/5/2019	6/20/2019
6	**Construction**	**31 days**	**6/24/2019**	**8/14/2019**
7	**Below Grade**	**13 days**	**6/24/2019**	**7/15/2019**
8	Grade Site	8 days	6/24/2019	7/4/2019
9	Set Foundation	9 days	6/24/2019	7/8/2019
10	Install Conduit	3 days	7/8/2019	7/10/2019
11	Dig Cable Trench	4 days	7/9/2019	7/15/2019
12	**Above Grade**	**20 days**	**7/11/2019**	**8/14/2019**
13	Erect Steel Structures	8 days	7/11/2019	7/24/2019
14	Install Equipment	6 days	7/16/2019	7/24/2019
15	Install Grounding	2 days	7/25/2019	7/29/2019
16	Install Bus and Jumpers	8 days	7/25/2019	8/7/2019
17	Lay Control Cable	12 days	7/25/2019	8/14/2019
18	**Fence**	**7 days**	**7/8/2019**	**7/17/2019**
19	Install Fence	7 days	7/8/2019	7/17/2019
20	**Site Restoration**	**23 days**	**7/16/2019**	**8/22/2019**
21	Remove Equipment	5 days	8/15/2019	8/22/2019
22	Lay Stoning	2 days	7/16/2019	7/17/2019
23	Lay Roadway	4 days	7/16/2019	7/22/2019
24	**Project Closeout**	**10 days**	**8/26/2019**	**9/10/2019**
25	Substantial Completion	10 days	8/26/2019	9/10/2019

Software interactions

- All described softwares are characterized by strong interactions!
- Most of time you will have to perform different parts of design, computation, simulation activities on different softwares and integrate results.
- Not often you will have availability of the most suitable software (cost is a major drive!) thus you might have to figure out creative solutions!

Software interactions

Which it the ADCS (Attitude Determination Control System) function?

A) Allow satellite orbital transfer
B) Determine and control the satellite orientation along the orbit
C) Determine and control the satellite and payload operative mode
D) Determine and control subsystem functionalities

What is PC104?

A) A satellite form factor
B) A high performant On-board computer for satellites
C) An electrical and mechanical standard for PCB design
D) A communication protocol for satellite telemetry

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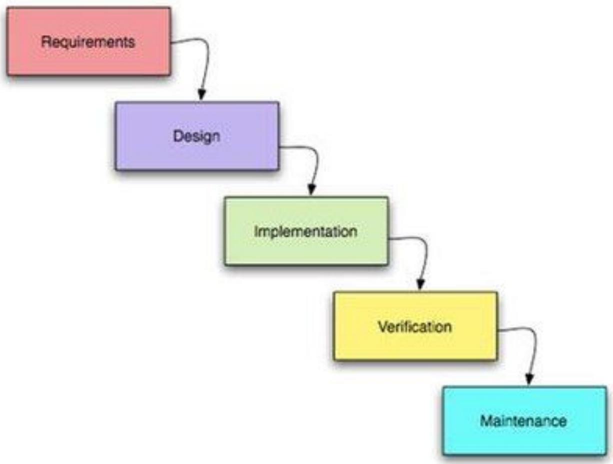
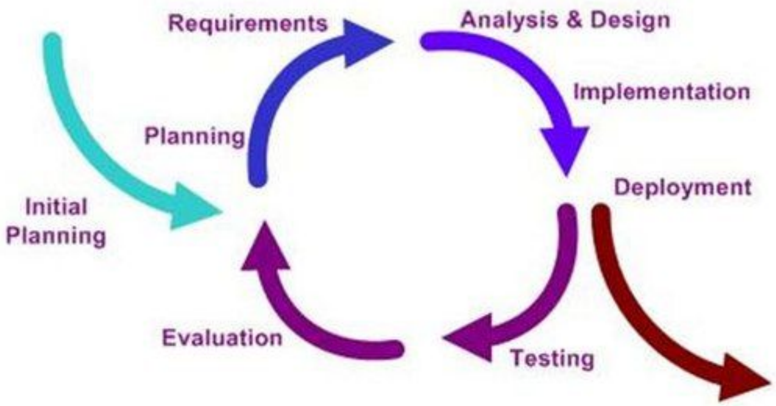
Iterative Design

Iterative design is a design methodology that consists in a rapid iteration of prototyping, testing and refinement in order to achieve the end requirements. At every iteration, the issues of a prototype are fixed and new ideas are gathered, so that another prototype could be manufactured. The process usually involve many different iterations. This is different from the more classical waterfall approach that you can see on the left in the figure below.



vs.

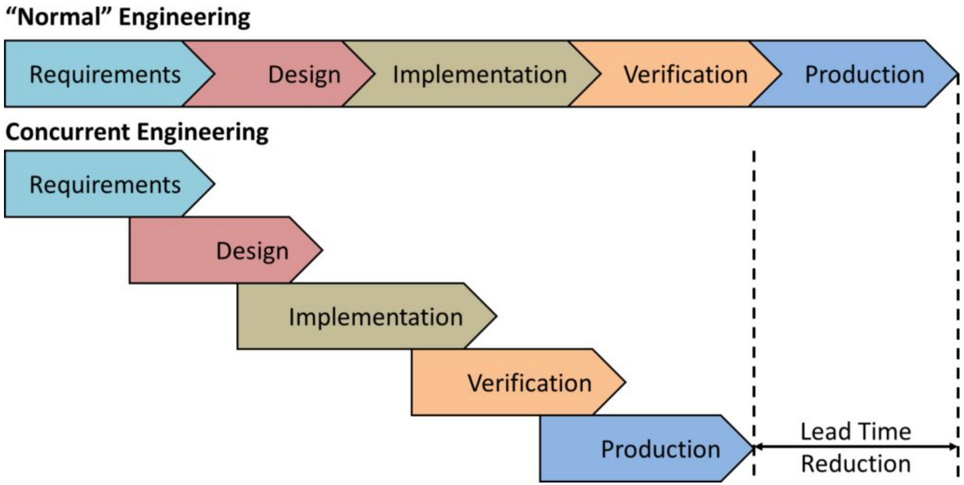


Iterative Design - Advantages

- The fast iteration can make much more clear eventual big misunderstandings and faulty core components of a design. Often times, ideas that seem brilliant on paper don't work as expected in real life, and with fast prototyping is much faster to fix these issues.
- The client could be more involved in the design phase, and thus it results in better communication between the client and the engineering team.
- Prototype are tested continuously, so that it is much easier to notice if the end requirements could be met.
- Fast prototyping allows the engineering team to investigate more potential design, given the velocity at which ideas could be adopted/discarded.
- Fast iteration has an effect on the team members, too: everyone is able to learn from every iteration and good practices are carried through the development of the design.

Concurrent Engineering

Satellite are a strong example of the practice of **Concurrent Engineering**. Tasks are often parallelized and done contemporarily, resulting in much faster development cycle and lower lead times.



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Concurrent Engineering and Iterative Design

Concurrent engineering and iterative design are almost always used in the Space Sector. As you have seen today, in a satellite the **choice of a particular subsystem affect the choice of every other one**. For example, a certain ADCS system will require a certain power to work, so maybe you will need a beefier Power System, and suddenly you don't have space available for the communication subsystem. Usually satellites are designed through iteration; in the first one, usually mass, volume and power budgets are based on common COTS, then the design is refined at every iteration until requirements are met. Furthermore, **every subsystem team will start working in parallel with each other** (*Concurrent Engineering*) and eventual inconsistencies between teams are found at the end of every iteration.

A Real World Example

The small satellites you will be integrating soon are a clear examples of these concepts, put in practice.

At the start of the project, every subsystem team worked on his own and we were able to have a finite design in a very small time. But that was only the first iteration! Then, every component has been tested, fixed and improved until the iteration you will work with.

Do you want to guess the number of iterations?